Marcus Vance

Senior Backend Engineer | marcus.vance@systems.dev

Professional Summary

Distributed systems and backend architect with 5+ years building scalable microservices, high-throughput APIs, and transactional database architectures.

Technical Skills & Technologies

Languages & Tools: Python, FastAPI, Go, PostgreSQL, Redis, Kafka, Docker, Kubernetes, SQLAlchemy, gRPC

Domain Competencies: Distributed Systems, Database Indexing & Concurrency, Event-Driven Systems, Caching Strategies, API Reliability

Professional Experience

• Lead Backend Engineer @ CloudSync Systems (2021 - Present): Designed distributed ingestion platform handling 45,000 requests/sec using FastAPI, Kafka, and Redis. Solved cache stampede using probabilistic early expiration and distributed mutex locks.

• Backend Engineer @ FinTech Core (2019 - 2021): Optimized PostgreSQL relational schema and B+ Tree indexes, reducing 99th percentile query latency from 850ms to 18ms. Implemented strict serializable transactions for ledger consistency.

• Systems Developer @ DataStream Inc (2018 - 2019): Built asynchronous worker services utilizing non-blocking event loops and circuit breaker patterns to guarantee 99.99% service uptime.

Education

B.S. in Software Engineering, Purdue University